CO4- ASSESMENT Tool

Comparitive Analysis

1) Linear vs Quadratic vs Double Hashing

Method	Primary clustering	Secondary clustering	cache Performance	Average Probes
Linear Probing	High	NO	Best	Higher as load increases
Quadratic Probing	NO	yes	Good	Lower than linear
Double hashing	No	NO	Good	Lowest

Best cache performance: Linear probing because it accesses consecutive memory locations, giving excellent spatia locality

Best overall collision resolution: Double hashing ~~Best~~ because it minimizes both primary and secondary clustering and generally gives fewer probes

2) Binary Heap vs Balanced BST

Operation	Binary Heap	Balanced BST
Insert	$O(\log n)$	$O(\log n)$
Extract-Min	$O(\log n)$	$O(\log n)$
Decrease-key	$O(\log n)$	$O(\log n)$
Arbitrary key update/search	$O(n)$	$O(\log n)$
Memory	$O(n)$, array-based	$O(n)$, extra pointers

Which is more efficient?

Binary heap can be stored in a simple array with no left/right-child pointers. A balanced BST requires extra memory for node pointers.

Final choice for scheduled

If the Scheduler mainly extract-min + decrease key a balanced BST is better when arbitrary key update/searches are frequent.

If memory efficiency and priority queue operations are the main concern, choose a binary heap.

2/8/26